COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/14/2026	**DATE SUBMITTED:**	10/14/2026
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ACTIVITY NO. 11

Containerization

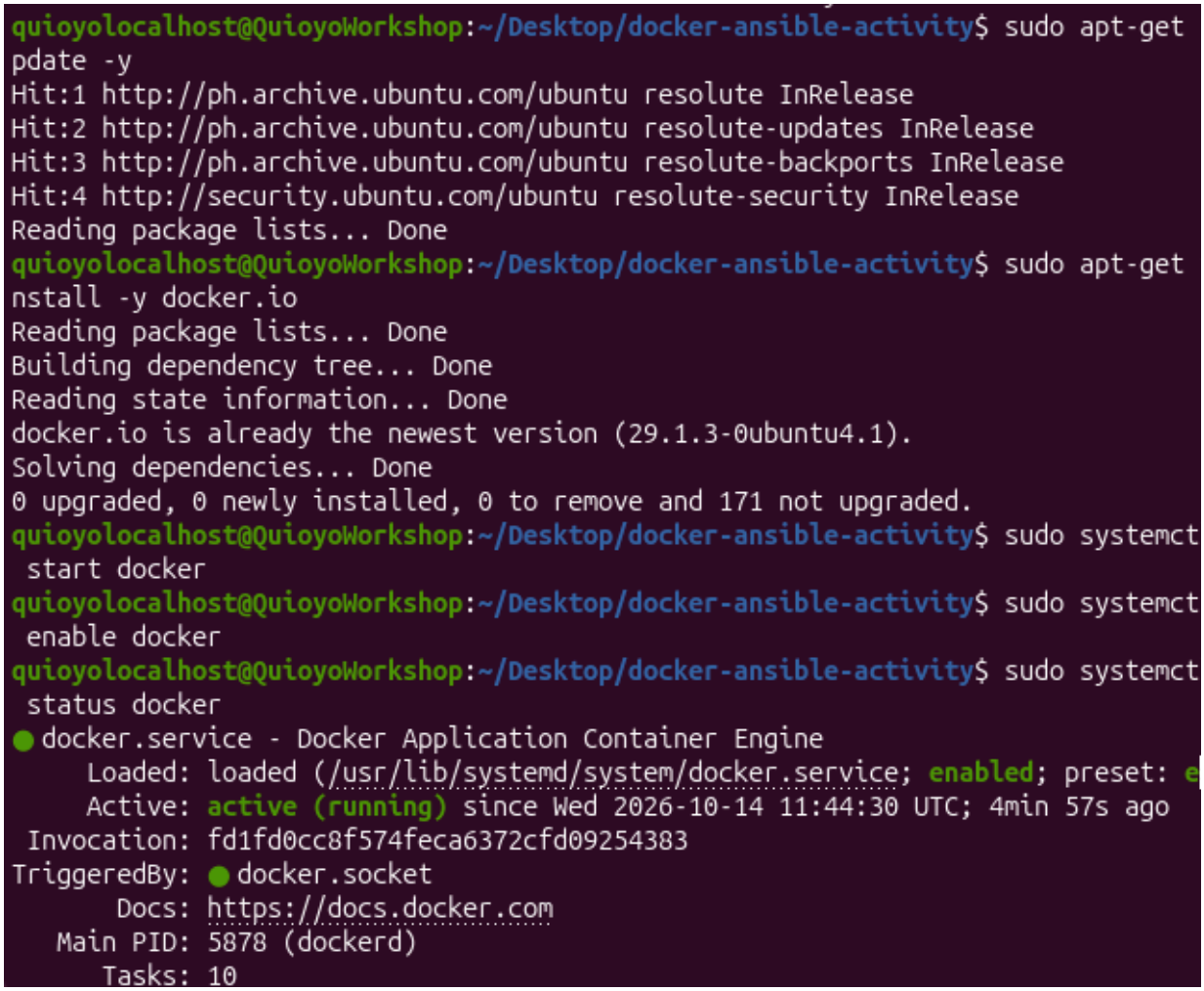
1. PROCEDURE OUTPUT

1. Create a new repository for this activity.

```
quioyolocalhost@QuioyoWorkshop:~/Desktop$ mkdir docker-ansible-activity
quioyolocalhost@QuioyoWorkshop:~/Desktop$ cd docker-ansible-activity
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$ git init
hint: Using 'master' as the name for the initial branch. This default branch nam
e
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/quioyolocalhost/Desktop/docker-ansible
-activity/.git/
```

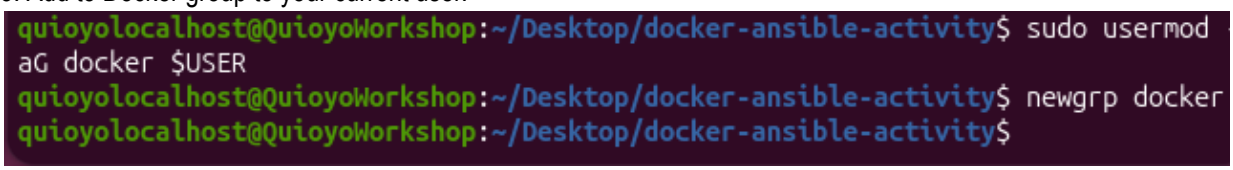
Explanation: I created a project directory named docker-ansible-activity using mkdir, moved into it with cd, and initialized a local Git repository using git init.

2. Install Docker and enable the docker socket.



Explanation: I updated package lists with apt-get update, installed Docker using apt-get install -y docker.io, and enabled and verified the Docker daemon service status using systemctl.

3. Add to Docker group to your current user.



Explanation: I added your active user account to the system docker security group using usermod -aG, then activated the updated group membership for your terminal session using newgrp docker.

4. Create a Dockerfile to install web and DB server.

```
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$ cat Dockerfile
cat << 'EOF' > Dockerfile
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && \
    apt-get install -y apache2 mariadb-server mariadb-client && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80 3306

CMD ["apachectl", "-D", "FOREGROUND]
EOF
```

Explanation: I verified the content of your Dockerfile using cat, which configures an Ubuntu 22.04 container image with Apache2 web server and MariaDB database server packages.

5. Install and build the Dockerfile using Ansible.

```
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$ sudo apt-get i
nstall -y ansible python3-pip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (13.1.0+dfsg-1ubuntu1).
ansible set to manually installed.
Solving dependencies... Done
Recommended packages:
  python3-dev
The following NEW packages will be installed:
  python3-pip python3-wheel
0 upgraded, 2 newly installed, 0 to remove and 171 not upgraded.
Need to get 1,444 kB of archives.
After this operation, 10.5 MB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-wheel
all 0.46.3-2 [27.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pip al
l 25.1.1+dfsg-1ubuntu2 [1,416 kB]
Fetched 1,444 kB in 2s (928 kB/s)
Selecting previously unselected package python3-wheel.
(Reading database… 213522 files and directories currently installed.)
Preparing to unpack …/python3-wheel_0.46.3-2_all.deb…
Unpacking python3-wheel (0.46.3-2)…
Selecting previously unselected package python3-pip.
Preparing to unpack …/python3-pip_25.1.1+dfsg-1ubuntu2_all.deb…
Unpacking python3-pip (25.1.1+dfsg-1ubuntu2)…
Setting up python3-wheel (0.46.3-2)…
Setting up python3-pip (25.1.1+dfsg-1ubuntu2)…
Processing triggers for man-db (2.13.1-1build1)…
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$
```

```
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$ ansible-playbo
ok -i inventory.ini playbook.yml
[WARNING]: Found both group and host with same name: localhost

PLAY [Build Docker Image and Run Web/DB Container] *****************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Host 'localhost' is using the discovered Python interpreter at '/usr/
bin/python3.14', but future installation of another Python interpreter could cau
se a different interpreter to be discovered. See https://docs.ansible.com/ansibl
e-core/2.20/reference_appendices/interpreter_discovery.html for more information
.
ok: [localhost]

TASK [Build Docker image from Dockerfile] **************************************
ok: [localhost]

TASK [Run Docker container] ****************************************************
ok: [localhost]

PLAY RECAP *********************************************************************
localhost                  : ok=3    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$
```

Explanation: I installed Ansible along with python3-pip using apt-get. Then, I executed your playbook using ansible-playbook -i inventory.ini playbook.yml, which successfully built the Docker image and started the container.

6. Add, commit and push it to your repository.

```
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$ git push -u o
igin main
Username for 'https://github.com': Angeloquioyo
Password for 'https://Angeloquioyo@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 851 bytes | 851.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Angeloquioyo/docker-ansible-activity..git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
quioyolocalhost@QuioyoWorkshop:~/Desktop/docker-ansible-activity$
```

Explanation: I pushed the committed local repository files and history to my GitHub remote repository using git push -u origin main.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:
Answer the following:
1. What are the benefits of implementing containerizations?
Containerization packages application code together with all its required dependencies and system libraries to guarantee consistent performance across development, testing, and production environments. It provides a lightweight, highly efficient alternative to traditional virtual machines by sharing the host operating system kernel to reduce resource overhead and accelerate boot times. Additionally, containerization enables seamless application isolation, rapid microservice scaling, and simple deployment portability across diverse cloud and on-premises infrastructure.

Conclusion: In this activity, I successfully established an automated Continuous Delivery workflow using Docker for containerization and Ansible as Infrastructure as Code (IaC). Prepared the environment by installing Docker, starting its socket daemon, and adding our user to the Docker group to grant non-root execution permissions. Next, I authored a custom Dockerfile to package both an Apache web server and a MariaDB database server into a single reusable container image. I then leveraged Ansible playbooks to automate the building of the Docker image and the deployment of the running containerized application on localhost. Finally, I initialized a local Git repository, committed all configuration files, and pushed the complete project to a remote GitHub repository.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance

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